

```
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime
```

```
# Task 1 -> Create Marks File
```

```
def create_marks_file():
```

```
    data = """Math,80
Science,75
English,90
Python,95"""
```

```
    with open("/tmp/student_marks.txt", "w") as file:
        file.write(data)
```

```
    print("Student marks file created successfully")
```

```
# Task 2 -> Read Marks File
```

```
def read_marks_file():
```

```
    with open("/tmp/student_marks.txt", "r") as file:
        records = file.readlines()
```

```
    print("Student Marks:")
```

```
for record in records:
```

```
    print(record.strip())
```

```
# Task 3 -> Calculate Total Marks
```

```
def calculate_total(ti):
```

```
    with open("/tmp/student_marks.txt", "r") as file:
```

```
        records = file.readlines()
```

```
    total_marks = 0
```

```
    for record in records:
```

```
        subject, marks = record.strip().split(",")
```

```
        total_marks += int(marks)
```

```
    print(f"Total Marks = {total_marks}")
```

```
    ti.xcom_push(key="total_marks", value=total_marks)
```

Bonus Task -> Percentage Calculation

```
def percentage_calculation(ti):
```

```
    total_marks = ti.xcom_pull(  
        task_ids="calculate_total",  
        key="total_marks"  
    )
```

```
    percentage = total_marks / 4
```

```
    print(f"Percentage = {percentage}")
```

```
    ti.xcom_push(key="percentage", value=percentage)
```

Task 4 -> Generate Result File

```
def generate_result(ti):
```

```
    total_marks = ti.xcom_pull(  
        task_ids="calculate_total",  
        key="total_marks"  
    )
```

```
    percentage = ti.xcom_pull(  
        task_ids="percentage_calculation",  
        key="percentage"  
    )
```

```
if percentage >= 40:
```

```
    result = "PASS"
```

```
else:
```

```
    result = "FAIL"
```

```
report = f"""
```

```
Student Result Summary
```

```
Total Marks = {total_marks}
```

```
Percentage = {percentage}
```

```
Result = {result}
```

```
"""
```

```
with open("/tmp/result.txt", "w") as file:
```

```
    file.write(report)
```

```
print("Result file generated successfully")
```

```
# DAG Definition
```

```
with DAG(
```

```
    dag_id="student_marks_workflow_dag",
```

```
    start_date=datetime(2025, 1, 1),
```

```
    schedule=None,
```

```
    catchup=False
```

```
) as dag:
```

```
# Create Tasks
```

```
task1 = PythonOperator(
```

```
    task_id="create_marks_file",
```

```
    python_callable=create_marks_file
```

```
)
```

```
task2 = PythonOperator(
```

```
    task_id="read_marks_file",
```

```
    python_callable=read_marks_file
```

```
)
```

```
task3 = PythonOperator(  
    task_id="calculate_total",  
    python_callable=calculate_total  
)
```

```
task4 = PythonOperator(  
    task_id="percentage_calculation",  
    python_callable=percentage_calculation  
)
```

```
task5 = PythonOperator(  
    task_id="generate_result",  
    python_callable=generate_result  
)
```

```
# Task Dependencies
```

```
task1 >> task2 >> task3 >> task4 >> task5
```